

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	176537	49.105166
1	[Git, Change, Create]	52105	14.493419
2	[Git, Change, Delete]	27843	7.744751
3	[Java, ByteCode, Member, Method]	13494	3.753463
4	[Java, ByteCode, Parameter]	13293	3.697553
5	[Git, Commit]	13215	3.675857
6	[File, Git]	11062	3.076983
7	[Git, Change, Rename]	10700	2.976290
8	[Java, ByteCode, Bound, ParameterizedType]	7279	2.024712
9	[Java, ByteCode, Bound]	7240	2.013863
10	[Java, ByteCode, Member, Field]	3609	1.003872
11	[Java, ByteCode, Bound, WildcardType]	2947	0.819731
12	[Java, Value, ByteCode, Annotation]	2931	0.815281
13	[Xml, Element]	2162	0.601377
14	[Java, ByteCode, Member, Method, Constructor]	2120	0.589695
15	[Xml, Text]	1450	0.403329
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.309033
17	[Java, ByteCode, Member, Method, Lambda]	968	0.269257
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.247004
19	[Type, File, Java, ByteCode, Class]	846	0.235322
20	[Json, Key]	702	0.195267
21	[Value, Json, Scalar]	685	0.190538
22	[Java, Value, ByteCode, Primitive]	672	0.186922
23	[Type, File, Java, ByteCode, JavaType]	649	0.180524
24	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.160775
25	[Type, File, Java, ByteCode, ExternalType]	411	0.114323
26	[Git, Change, Copy]	334	0.092905
27	[Author, Git, Person]	299	0.083169
28	[Value, Array]	287	0.079831
29	[Committer, Git, Person]	252	0.070096

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

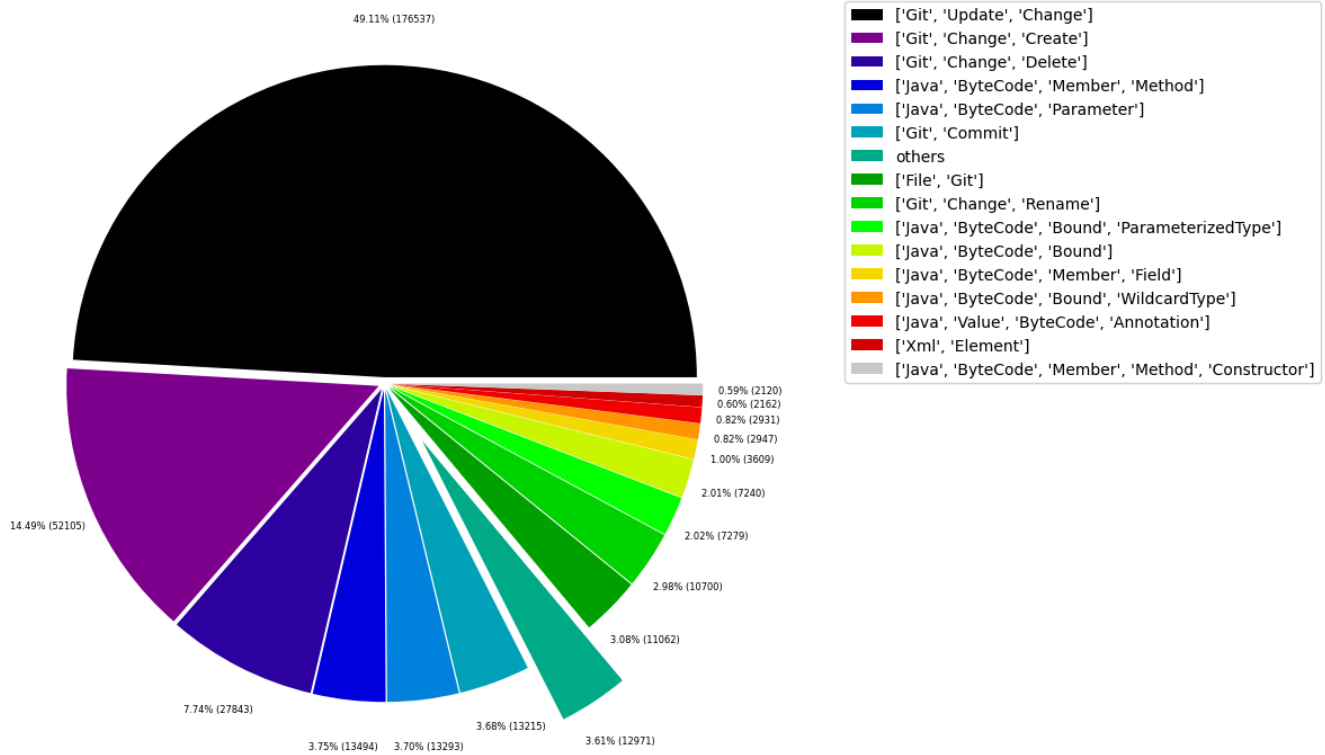


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000278
1	[Package, File, Json, NPM]	1	0.000278
2	[Repository, File, Git]	1	0.000278
3	[File, TS, Scan]	1	0.000278
4	[File, Json]	2	0.000556
5	[File]	3	0.000834
6	[Java, ByteCode, Member, Method, GenericDeclar...	4	0.001113
7	[Maven, Exclusion]	5	0.001391
8	[Value, Array, Json]	6	0.001669
9	[Dependency, NPM]	7	0.001947
10	[Type, File, Java, ByteCode, Void]	9	0.002503
11	[File, Maven, Xml, Pom, Document]	9	0.002503
12	[Java, ManifestSection]	9	0.002503
13	[File, Java, Manifest]	9	0.002503
14	[Artifact, File, Jar, Archive, Zip, Java]	9	0.002503
15	[File, Java, ServiceLoader]	10	0.002782
16	[File, Java, Properties]	12	0.003338
17	[Maven, ExecutionGoal]	16	0.004451
18	[Maven, PluginExecution]	16	0.004451
19	[Xml, Attribute]	18	0.005007
20	[jQAssistant, Rule, Concept]	19	0.005285
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005841
22	[Maven, Configuration]	21	0.005841
23	[Maven, Plugin]	21	0.005841
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006119
25	[Type, File, Java, ByteCode, Enum]	28	0.007788
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008345
27	[Xml, Namespace]	36	0.010014
28	[Type, File, Java, ByteCode, Annotation]	44	0.012239
29	[File, Directory]	48	0.013352

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

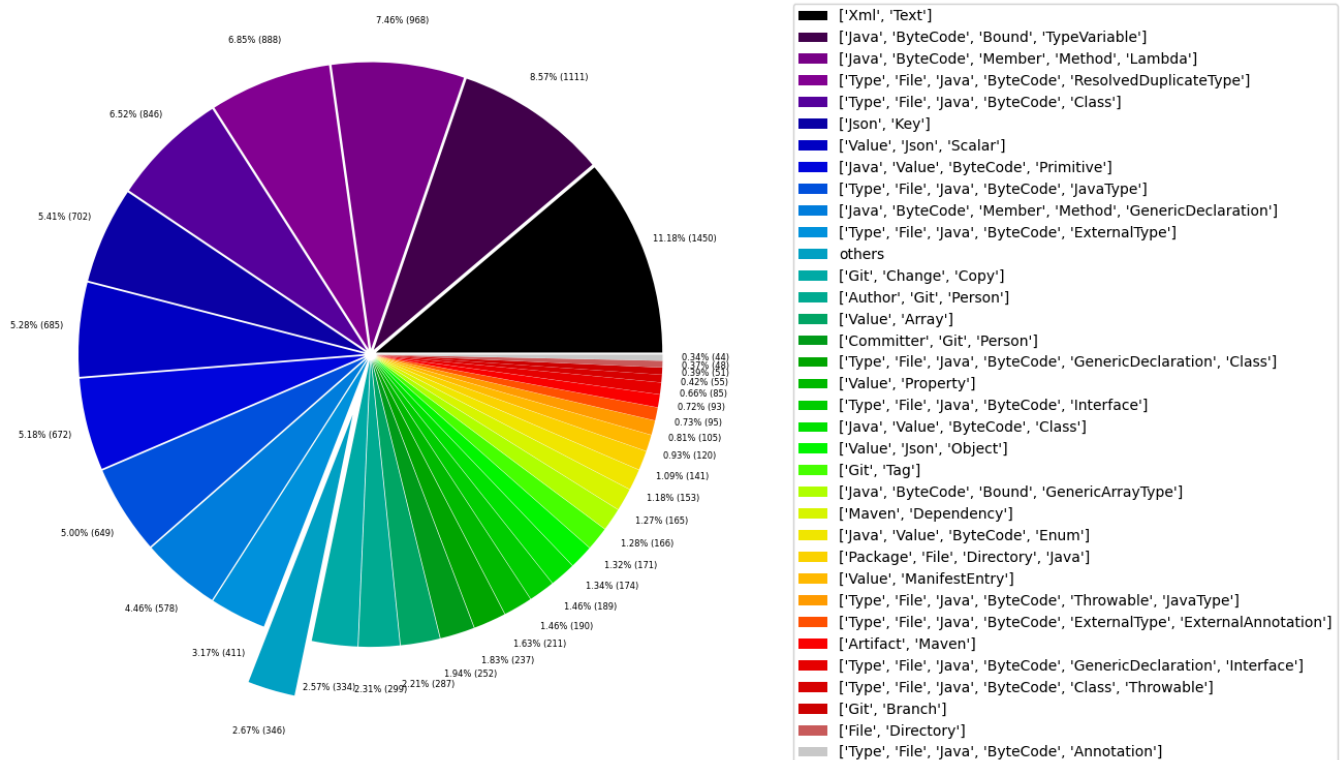


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

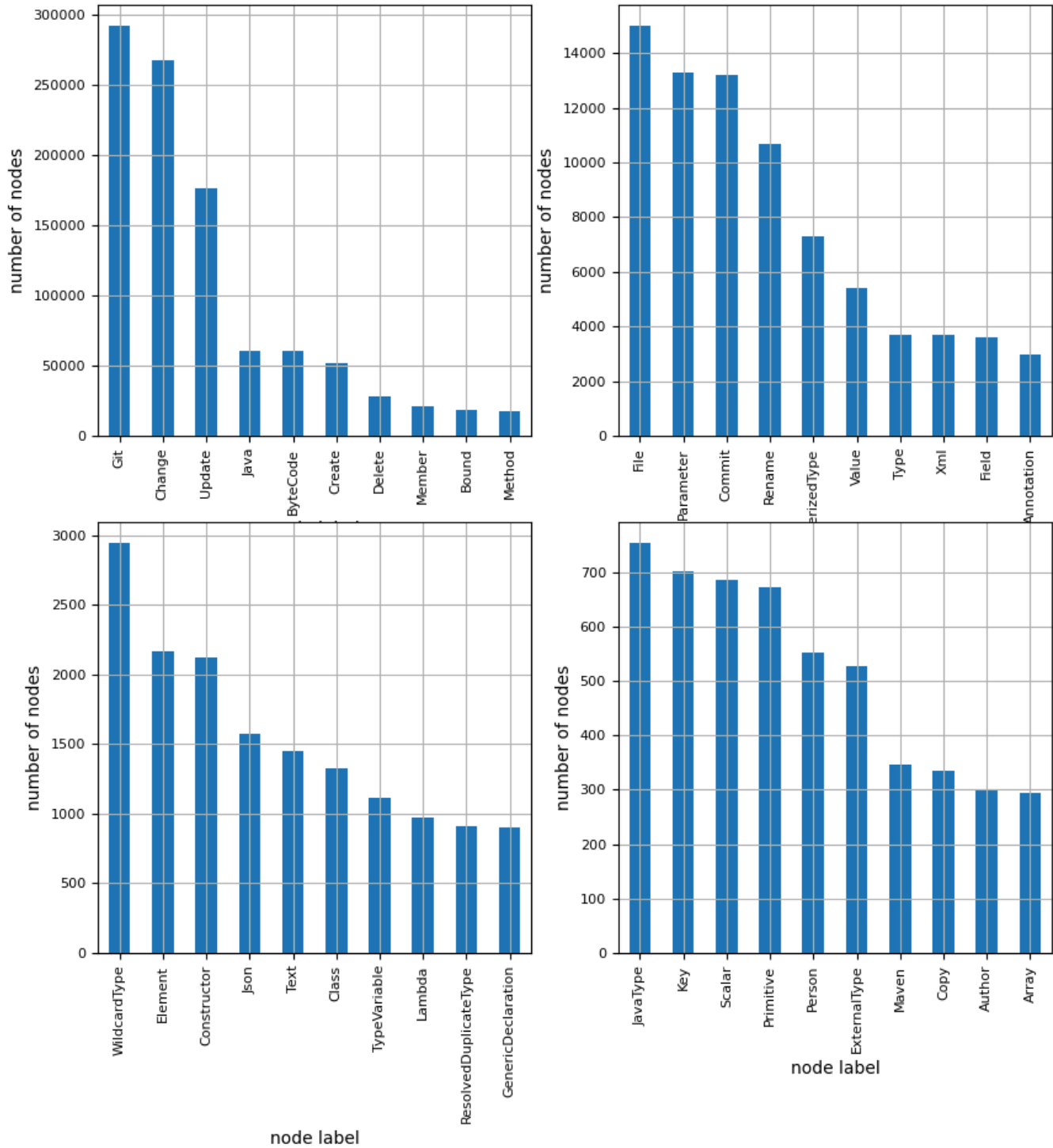
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	292570	81.380665
1	Change	267519	74.412530
2	Update	176537	49.105166
3	Java	60659	16.872782
4	ByteCode	60469	16.819932
5	Create	52105	14.493419
6	Delete	27843	7.744751
7	Member	20773	5.778175
8	Bound	18743	5.213514
9	Method	17164	4.774303
10	File	15023	4.178767
11	Parameter	13293	3.697553
12	Commit	13215	3.675857
13	Rename	10700	2.976290
14	ParameterizedType	7279	2.024712
15	Value	5428	1.509841
16	Type	3715	1.033357
17	Xml	3675	1.022230
18	Field	3609	1.003872
19	Annotation	2975	0.827520
20	WildcardType	2947	0.819731
21	Element	2162	0.601377
22	Constructor	2124	0.590807
23	Json	1570	0.436708
24	Text	1450	0.403329
25	Class	1327	0.369116
26	TypeVariable	1111	0.309033
27	Lambda	968	0.269257
28	ResolvedDuplicateType	910	0.253124
29	GenericDeclaration	904	0.251455
30	JavaType	754	0.209731
31	Key	702	0.195267
32	Scalar	685	0.190538
33	Primitive	672	0.186922
34	Person	551	0.153265
35	ExternalType	527	0.146589
36	Maven	346	0.096243
37	Copy	334	0.092905
38	Author	299	0.083169
39	Array	293	0.081500

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1115601

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	267519	23.979810
1	MODIFIES	267519	23.979810
2	UPDATES	176537	15.824385
3	CREATES	63139	5.659640
4	DELETES	38543	3.454909
5	INVOKES	36776	3.296519
6	COMMITTED	26430	2.369127
7	DEPENDS_ON	22492	2.016133
8	OF_TYPE	21920	1.964860
9	DECLARES	21234	1.903369
10	OF_RAW_TYPE	17353	1.555484
11	HAS_PARENT	15980	1.432412
12	HAS	14429	1.293384
13	HAS_COMMIT	13215	1.184563
14	RETURNS	12844	1.151308
15	HAS_FILE	11062	0.991573
16	RENAMES	10700	0.959124
17	READS	9378	0.840623
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.753585
19	HAS_NEW_NAME	6299	0.564628
20	OF_GENERIC_TYPE	5996	0.537468
21	RESOLVES_TO	5270	0.472391
22	SIMILAR	4078	0.365543
23	WRITES	3944	0.353531
24	CONTAINS	3936	0.352814
25	RETURNS_GENERIC	3593	0.322069
26	ANNOTATED_BY	2919	0.261653
27	REQUIRES	2230	0.199892
28	HAS_FIRST_CHILD	2162	0.193797
29	HAS_LAST_CHILD	2162	0.193797

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

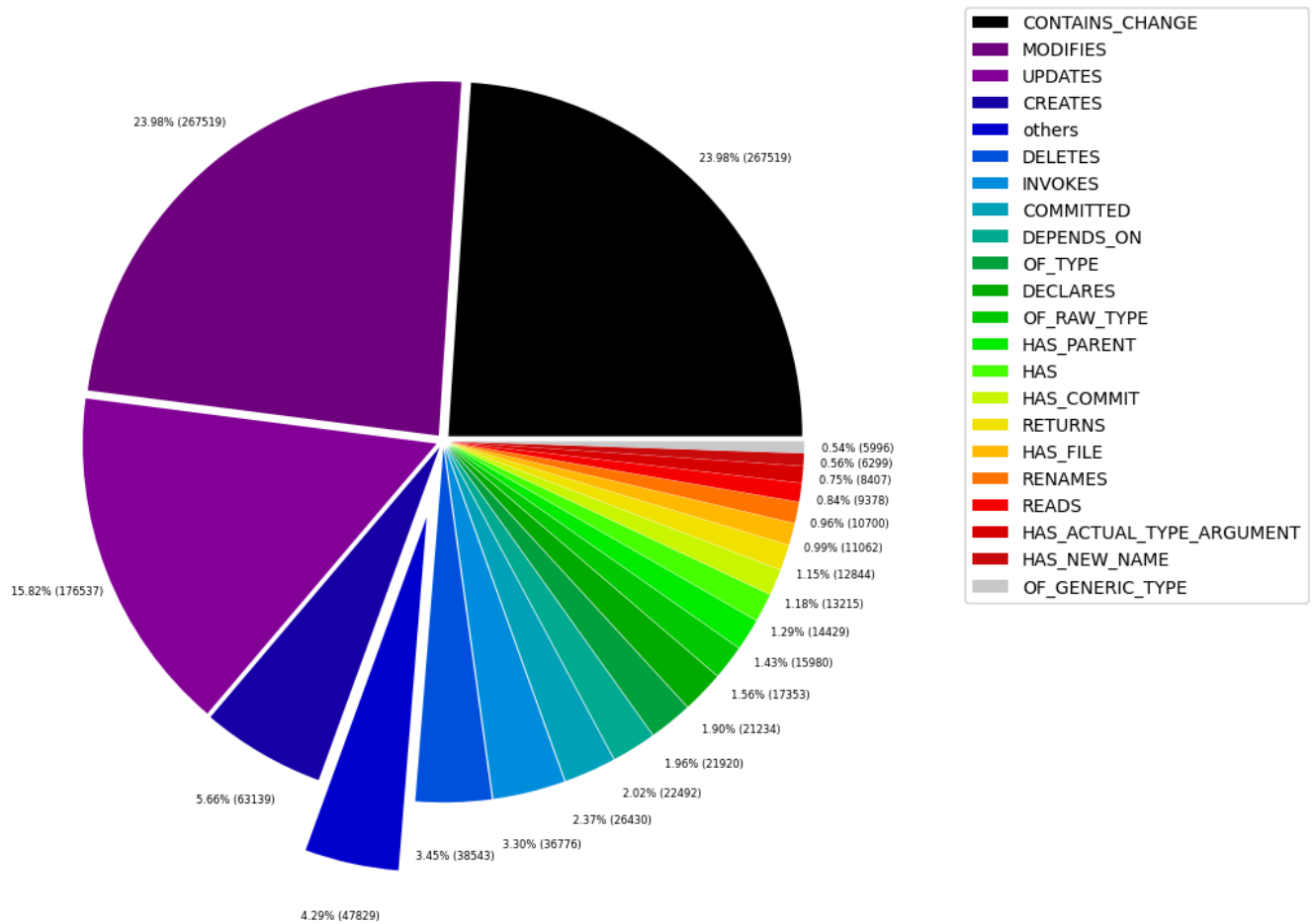


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

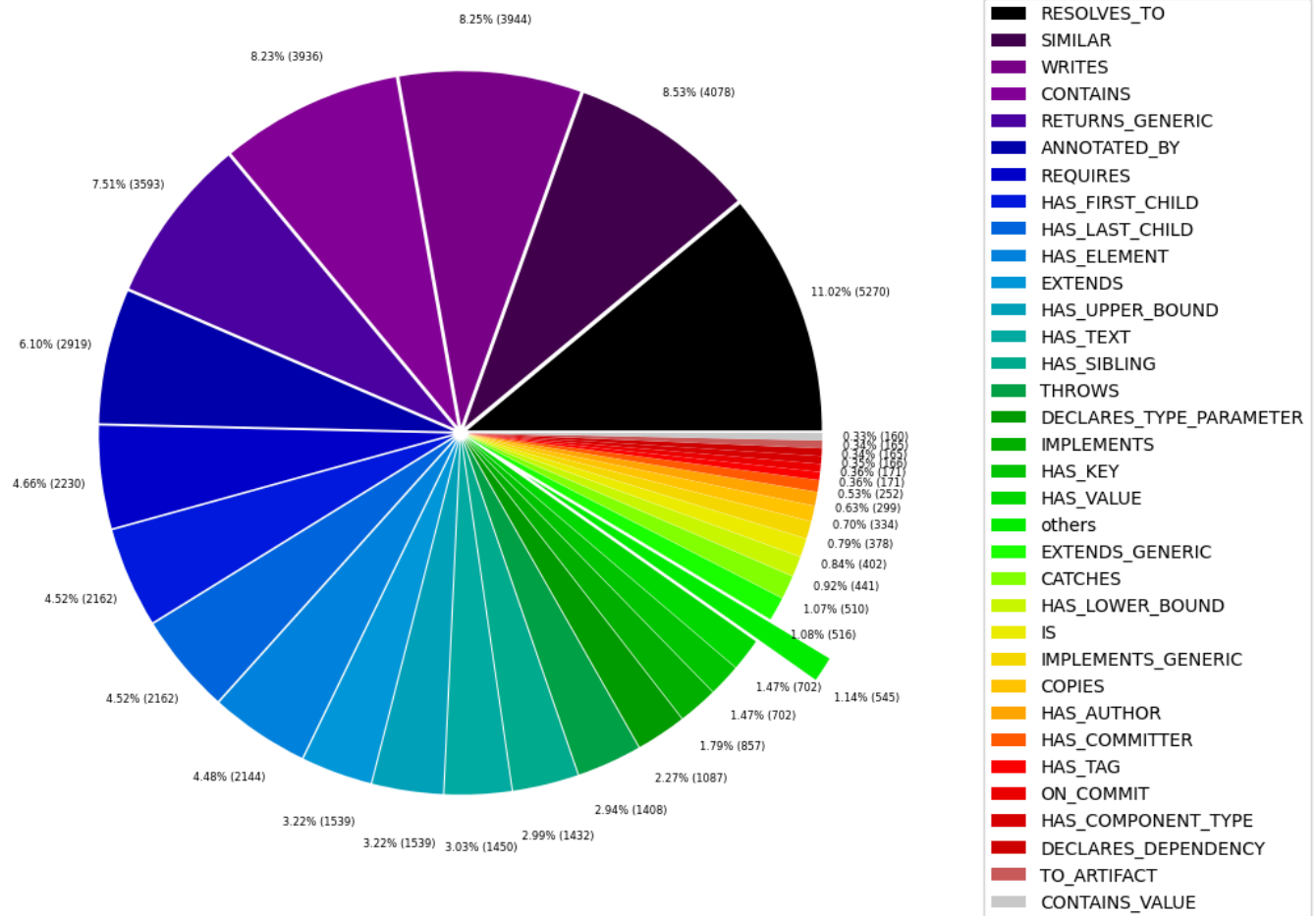
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000090
1	THROWS_GENERIC	5	0.000448
2	EXCLUDES	5	0.000448
3	DECLARES_DEV_DEPENDENCY	7	0.000627
4	DESCRIBES	9	0.000807
5	HAS_ROOT_ELEMENT	11	0.000986
6	HAS_GOAL	16	0.001434
7	HAS_EXECUTION	16	0.001434
8	OF_NAMESPACE	18	0.001613
9	HAS_ATTRIBUTE	18	0.001613
10	INCLUDES_CONCEPT	19	0.001703
11	IS_ARTIFACT	21	0.001882
12	HAS_CONFIGURATION	21	0.001882
13	USES_PLUGIN	21	0.001882
14	REQUIRES_TYPE_PARAMETER	24	0.002151
15	REQUIRES_CONCEPT	28	0.002510
16	DECLARES_NAMESPACE	36	0.003227
17	HAS_DEFAULT	39	0.003496
18	HAS_BRANCH	51	0.004572
19	HAS_HEAD	52	0.004661
20	COPY_OF	127	0.011384
21	CONTAINS_VALUE	160	0.014342
22	DECLARES_DEPENDENCY	165	0.014790
23	TO_ARTIFACT	165	0.014790
24	HAS_COMPONENT_TYPE	166	0.014880
25	ON_COMMIT	171	0.015328
26	HAS_TAG	171	0.015328
27	HAS_COMMITTER	252	0.022589
28	HAS_AUTHOR	299	0.026802
29	COPIES	334	0.029939

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	UPDATES	[File, Git]	176537	176537	
1	[Git, Update, Change]	MODIFIES	[File, Git]	176537	176537	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	176537	13215	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	52105	13215	
4	[Git, Change, Create]	MODIFIES	[File, Git]	52105	52105	
5	[Git, Change, Create]	CREATES	[File, Git]	52105	52105	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	27843	13215	
7	[Git, Change, Delete]	DELETES	[File, Git]	27843	27843	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	27843	27843	
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22590	13404	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	15971	13215	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13215	1	
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13215	252	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	13215	299	
14	[Repository, File, Git]	HAS_FILE	[File, Git]	11062	1	
15	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10700	13215	
16	[Git, Change, Rename]	DELETES	[File, Git]	10700	10700	
17	[Git, Change, Rename]	CREATES	[File, Git]	10700	10700	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	10700	10700	
19	[Git, Change, Rename]	RENAMES	[File, Git]	10700	10700	
20	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8517	13404	
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8446	13404	
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6299	11062	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695	13293	
28	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2519	2120	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2474	7279	

Graph Density

```
total_number_of_nodes (vertices): 359508
total_number_of_relationships (edges): 1115601
-> total directed graph density: 8.631633360611198e-06
-> total directed graph density in percent: 0.0008631633360611198
```